Prise en charge percutanée des tumeurs desmoïdes : actualisation 2026 des indications et thérapies ablatives

Titre original : Percutaneous Management of Desmoid Tumors

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1. Pertinence Clinique & Lien avec l'Agenda Interventionnel (Cochin)

Mise à jour 2026 intégrée aux discussions de RCP Sarcomes / Tumeurs des parties molles (Cochin) pour positionner la cryoablation face aux inhibiteurs de gamma-sécrétase et de tyrosine kinase.

2. Synthèse Clinique en Français

Résumé exécutif : Revue récente (*Current Oncology Reports*, 2026) précisant la place de la radiologie interventionnelle (cryoablation, micro-ondes, HIFU, embolisation transartérielle) dans l'algorithme moderne des tumeurs desmoïdes aux côtés de la surveillance active et des nouvelles thérapies ciblées (nirogacestat, sorafénib).

Analyse détaillée : Avec l'évolution des recommandations internationales du Desmoid Tumor Working Group (DTWG), la chirurgie d'exérèse a laissé place à une approche conservatrice. Cet article de 2026 analyse les résultats à long terme de la cryoablation percutanée en première ou deuxième ligne pour les desmoïdes extra-abdominaux symptomatiques ou progressifs (ceinture scapulaire, creux sus-claviculaire, paroi thoracique, membres). La cryoablation permet d'éviter la toxicité chronique des traitements systémiques au long cours tout en préservant l'option d'un retraitement focal en cas de reliquat évolutif.

3. Protocole Technique & Paramètres Opératoires

Discussion systématique en RCP Sarcomes • Ablation complète (A0) visée lorsque l'anatomie le permet ; cryoablation de réduction volumétrique ('debulking' antalgique) avec épargne nerveuse lorsque la tumeur engaine le plexus brachial ou sciatique.

4. Points Clés à Retenir pour la Pratique

- Taux de réponse objective de 55 à 80 % et contrôle symptomatique > 85 % après cryoablation des tumeurs desmoïdes.
- En cas de contiguïté avec le plexus brachial (creux sus-claviculaire), une cryoablation subtotale planifiée soulage les douleurs sans déficit moteur.
- Faible morbidité comparée à la chirurgie et absence de toxicité systémique prolongée (contrairement aux ITK ou inhibiteurs de gamma-sécrétase).
- Possibilité de séances itératives de cryoablation sur un bourgeon résiduel détecté sur l'IRM de suivi.

5. Original PubMed Abstract (Texte Intégral de l'Abstract)

Purpose of review To provide a comprehensive review of percutaneous image-guided techniques in the management of desmoid tumors including evolving thermal and non-thermal ablation options, as well as trans-arterial catheter options. Beyond each treatment modality, we aim to highlight considerations related to patient selection, technical components, and the growing interplay between percutaneous therapies and emerging systemic therapies. Recent findings Percutaneous ablation, particularly cryoablation, has emerged as a safe and effective treatment for extra-abdominal desmoid tumors, with prospective data demonstrating durable disease control and significant improvements in pain and functional status. A meta-analysis of 694

desmoid tumors treated with thermal ablation reported pooled local control of 88%, symptom relief in 90% of patients, and major complication rates below 5%. Transarterial doxorubicin-eluting embolization has shown promise in both adult and pediatric settings, while bland particle and radioembolic approaches remain under investigation. Irreversible electroporation is emerging as a non-thermal alternative near critical structures. The National Comprehensive Cancer Network (NCCN) Soft Tissue Sarcoma Guidelines now formally incorporate cryoablation along with embolization into the treatment algorithm for progressive, morbid, or symptomatic extra-abdominal desmoid tumors, marking a significant milestone in the recognition of percutaneous techniques as standard-of-care options. Percutaneous image-guided techniques now offer local tumor control with rates similar to surgical resection and a favorable safety profile. Some of these percutaneous techniques are endorsed alongside systemic therapy and radiation as primary options for progressive or symptomatic extra-abdominal desmoid tumors per various guidelines. An important consideration involves the level of data, comprised predominantly of retrospective data, without direct comparative date. The ongoing prospective CRYODESMO-02 trial will help define optimal sequencing and strengthen the evidence base underpinning guideline recommendations.

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6. Full-Text Sections (Europe PMC Open Access JATS XML)

Introduction

Desmoid tumors are rare clonal neoplasms characterized by local invasiveness without metastatic potential, with an annual incidence of approximately 3–4 cases per million in the U.S. population [1]. They typically affect younger patients, show a female predominance, and can arise at virtually any anatomic site [2]. The clinical course is highly variable, ranging from spontaneous regression to persistent local progression causing significant pain, functional impairment, and diminished quality of life [3, 4].

Management has undergone fundamental transformation over the past two decades. Practice initially mirrored that used for soft tissue sarcomas, but recognition that margin positivity exceeds 40%, that 5-year recurrence-free survival after resection is only approximately 53%, and that margin status does not consistently predict recurrence [5, 6], together with observations that 20–30% of tumors spontaneously regress [7, 8], prompted a shift toward watchful waiting. A multicenter EMSOS study of 388 patients confirmed comparable outcomes between surgery (36.6% recurrence) and active surveillance (40.2% progression) and identified pain as a novel predictor of progression during observation [9]. Active surveillance is now the preferred initial strategy for minimally symptomatic or non-critically located disease.

When active treatment is warranted, the therapeutic landscape now includes systemic agents (tyrosine kinase inhibitors, gamma-secretase inhibitors), radiation, surgery, and a rapidly expanding portfolio of percutaneous image-guided techniques [10–12]. A critical milestone in this evolution was the incorporation of ablation and embolization into the NCCN Soft Tissue Sarcoma Guidelines as recognized local therapy options for desmoid tumors [13]. Both percutaneous treatment options listed alongside systemic therapy and definitive radiation as options for progressive, morbid, or symptomatic extra-abdominal desmoid tumors - abdominal wall, pelvis, trunk/extremity, head/neck, and intrathoracic locations - with surgery positioned as the less-preferred alternative [13]. DEB-TACE is specifically recognized as a reasonable alternative for selected extra-abdominal desmoid tumors when cryoablation is not technically feasible. The Desmoid Tumor Working Group (DTWG) 2024 consensus similarly recognizes local ablative techniques as valid treatment options [14].

This review provides a comprehensive update on percutaneous management of desmoid tumors including cryoablation, radiofrequency ablation, microwave ablation, irreversible electroporation, chemical ablation, high-intensity focused ultrasound, and transarterial embolization, with emphasis on efficacy, safety, patient selection, and the integration of these techniques with systemic therapy.

Image-guided ablation achieves tissue destruction through the delivery of extreme temperatures, electrical fields, acoustic energy, or cytotoxic agents. Ultrasound, CT, and MRI provide the basis for procedural planning, probe placement, real-time monitoring, and post-treatment assessment, enabling targeted tumor destruction

while sparing adjacent critical structures [15]. The expanding role of interventional radiology in desmoid tumor care, and the full armamentarium of minimally invasive options, has been reviewed recently [16].

Each modality occupies a distinct niche. Chemical ablation, the earliest described percutaneous technique for desmoid tumors, uses direct intratumoral injection of cytotoxic agents such as acetic acid, inducing coagulative necrosis through protein denaturation and collagen dissolution [10]. Clark reported CT-guided intratumoral injection of 50% acetic acid in two patients with refractory disease, achieving complete and partial responses with durable control at 24–36 months. Though largely superseded by thermal modalities with more predictable ablation zones, this work established the framework for percutaneous desmoid management [10]. Cryoablation, uses the expansion of pressurized argon to generate lethal cold temperatures via the Joule-Thomson effect. Radiofrequency ablation (RFA) induces coagulative necrosis through frictional heating from an alternating electric current delivered via a percutaneous electrode [17]. Microwave ablation (MWA) uses electromagnetic energy to generate heating, offering shorter procedure times and reduced susceptibility to heat-sink effects compared with RFA [18]. Irreversible electroporation (IRE), a non-thermal modality, uses brief high-voltage electrical pulses to induce cell membrane permeabilization and apoptosis while better preserving extracellular matrix and neurovascular structures. High-intensity focused ultrasound (HIFU) is non-invasive, using focused acoustic energy to heat tissues to ablative temperatures under MRI or ultrasound guidance [19]. Transarterial embolization uses the typical hypervascularity of desmoid tumors, delivering cytotoxic drugs, bland particles, or radionuclides to induce ischemia, necrosis, and, where applicable, local drug exposure [20].

RFA was among the earliest thermal modalities applied to desmoid tumors, with case reports demonstrating feasibility for both abdominal wall and intra-abdominal tumors in familial adenomatous polyposis with multiple sessions [17, 21, 22]. RFA has technical limitations in desmoid tumors: reduced efficacy against dense collagenous stroma due to high tissue impedance, greater risk of thermal injury compared with cryoablation, and susceptibility to heat-sink effects. Cryoablation has largely superseded RFA for desmoid treatment, although RFA remains viable where cryoablation is unavailable.

Desmoid-specific MWA data remain relatively sparse. A retrospective study of 9 patients (mean diameter 10.4 cm) treated with MWA reported 70.4% mean volume reduction, complete response in two patients, and functional improvement in eight of nine [18]. Zhang et al. recently reported the first pediatric experience, treating 8 children (mean age 9.2 years) with 10 recurrent desmoid tumors across 11 ultrasound-guided MWA sessions; mean volume reduction was 55.86% with an 80% overall effectiveness rate, significant pain reduction, and only one grade 1 peroneal nerve injury that recovered [23].

Limitations

With the exception of CRYODESMO-01 [24, 25], the current literature consists of retrospective single-center series and pooled analysis of heterogenous studies. No randomized trial has compared percutaneous ablation with systemic therapy, surgery, or surveillance, and referral and selection bias likely report improved efficacy. Follow-up durations are often inadequate for a disease characterized by late recurrence.

Technical limitations also warrant emphasis. Durable control depends on complete tumor coverage, which becomes harder as size increases; largest diameter was the only predictor of failure in CRYODESMO-01, and recurrences cluster more commonly at the periphery. Large or anatomically complex lesions may require multiple probes or staged treatment sessions, and tumors near bowel, skin, or neurovascular structures remain challenging despite thermoprotective techniques. Although major complication rates stay below 5%, procedure related morbidity is not negligible, and some patients may need repeat ablation or transition to alternative therapy.

Conclusions

Percutaneous image-guided ablation is now integral to desmoid tumor management, with prospective and meta-analytic data demonstrating local control comparable to surgery, high rates of symptom relief, and major complication rates below 5%. The inclusion of ablation and embolization in the NCCN Soft Tissue Sarcoma Guidelines Version 3.2026, with cryoablation identified as the most commonly utilized modality and DEB-TACE

recognized when cryoablation is not technically feasible [13], represents a significant milestone for the field. Patient selection requires multidisciplinary evaluation weighing tumor characteristics, location, symptom burden, and reproductive considerations. The CRYODESMO-02 trial, together with prospective studies evaluating combination strategies such as embolization followed by cryoablation and nirogacestat combined with cryoablation, will refine optimal sequencing, patient selection, and the potential systemic-therapy-sparing role of local intervention.

Key References

Kurtz JE et al., 2021. CRYODESMO-O1: A prospective, open phase II study of cryoablation in desmoid tumour patients progressing after medical treatment. European Journal of Cancer.■ This multicenter French phase II trial (NCT02476305) of 50 patients with non-abdominopelvic desmoid tumors progressing after at least two lines of medical therapy met its primary endpoint with an 86% non-progression rate at 12 months and median PFS not reached at 31 months of follow-up. Cryoablation was safe (22% grade 3–4 events, no grade 5) and produced durable improvements in Brief Pain Inventory and EQ5D-3L scores, establishing the first prospective evidence for cryoablation as a tissue-sparing local therapy in progressive, heavily pretreated extra-abdominal disease.Gounder M et al., 2023. Nirogacestat, a γ-Secretase Inhibitor for Desmoid Tumors. New England Journal of Medicine.■ The phase III DeFi trial (NCT03785964) randomized 142 adults with progressing desmoid tumors to oral nirogacestat 150 mg twice daily or placebo and demonstrated a 71% reduction in the risk of progression or death (HR 0.29; 95% CI 0.15–0.55; P<0.001), with objective response rates of 41% vs. 8% and significant improvements across all patient-reported outcomes. Despite frequent but mostly low-grade toxicities (diarrhea 84%, ovarian dysfunction in 75% of women of childbearing potential), this trial provided the first high-level randomized evidence supporting a targeted systemic therapy and led to the first FDA-approved treatment for desmoid tumors.Huang K et al., 2023. Efficacy and safety of different thermal ablative therapies for desmoid-type fibromatosis: a systematic review and meta-analysis. Quantitative Imaging in Medicine and Surgery.■ This meta-analysis of 23 studies and 568 patients with 694 desmoid tumors treated by cryoablation, HIFU, or microwave ablation reported pooled rates of 90% symptom relief, 88% local control, 76% non-perfused volume, and only 3% major complications. Although HIFU showed numerically higher local control (99%) than cryoablation (80%) or microwave ablation (78%) (P=0.01), this difference likely reflects selection bias toward superficial tumors with adequate acoustic windows rather than intrinsic modality superiority, as HIFU cohorts systematically exclude deep, bone-adjacent, or gas-shielded lesions that are routinely treated with cryoablation. By quantitatively aggregating fragmented retrospective data across modalities and imaging guidance, this analysis supports the integration of image-guided thermal ablation into multidisciplinary management of progressive or symptomatic desmoid disease and helps reconcile divergent NCCN, ESMO, and Desmoid Tumor Working Group recommendations.

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